# Tan Yu Min

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## Professional Summary

Computer Science undergraduate at the Singapore University of Technology and Design with experience building full-stack web applications using React, Typescript, Go, and modern web technologies. Experienced in leading student engineer teams, developing production software, and building developer-focused tooling with an interest in scalable systems and developer experience.

## Core Skills

- Languages: Javascript/TypeScript, Go, C++, C#, Python, Java, SQL
- Full-stack: Next.js, ASP.NET
- Frontend: React, HTML, CSS, Elm
- Backend: Express, Gorilla Mux
- Infrastructure: Docker, Linux, Git, AWS, Supabase, Vercel
- Test Frameworks: Jest, Vitest
- Other Interests: Keyboards, Coffee, Basketball

## Professional Experience

**Sign Pte Ltd** Singapore
*Software Engineer Intern* March 2021 - August 2021

- Worked with a Linux development environment, using the CLI to install packages to set up the dev environment, and editing files using Vim.
- Gathered and clarified requirements from superiors to ensure that features implemented met the user and business needs.
- On the frontend:
  - Implemented features on Elm frontend, using functional programming paradigms like recursion and higher order functions.
  - Implemented responsive UI, ensuring adequate margins/padding, responsive layouts, as well as a design consistent with the rest of the application for various viewports.
  - Implemented new UX features such as loading skeletons to improve perceived application responsiveness.
- On the backend:
  - Wrote and updated RESTful API endpoints using Express
  - Developed passing and failing Jest unit tests to ensure proper API functionality before deployment.
- Passed strict code review sessions with supervisors before deployment, ensuring principles such as SRP, YAGNI and KISS were upheld to the team's standards.

## Academic Projects

**SUTD ROOT Student Government** root.sutd.edu.sg
*Technical Lead / Secretary of ROOTech* Mar 2026 - Present

- Currently leading a team of 10 developers to build and maintain internal and external applications for ROOT Student Government and the general student population respectively.
- Liaise with key stakeholders such as ROOT Cabinet and School Offices to gather user/business requirements.
- Plan and assign tasks to developers, and assist with any blockages during development process.
- Review and merge developers' code into production, and request changes if necessary.
- Improved developer workflows and reduced infrastructure costs by evaluating and introducing tools such as SOPS for secret management and migrating backend services from DigitalOcean to Supabase.
- Wrote new documentation and cleaned up old documentation to facilitate the onboarding process for new developers.

**ETI Microservice Assignment** *Docker, Docker Compose, SSH, PuTTY, Microservices*

- Built the ClassExchange Microservice and Microfrontend as part of a network of 20+ microservices
- Liaised with developers of Class and Authentication Microservices and consumed their APIs to allow for class exchange functionality.
- Built Backend RESTful API with Go and Gorilla Mux, and used MySQL to store class exchange data.
- Containerised Applications with Docker, and used Docker Compose to orchestrate the microservices, allowing for easy deployment and testing.
- Used SSH and PuTTY to connect to the remote server, and successfully deployed the microservices onto the server for testing.

**LeafMeBe** *ESP32 Microcontroller, Arduino, C++*

- Developed automatic plant-watering system using an ESP32 microcontroller and Arduino ESP Core.
- Interfaced moisture sensors and water pumps via GPIO control to automate watering cycles and eliminate manual watering
- Awarded **Top in Cohort for Innovativeness and Risk-Taking**

## Personal Projects

**CIS Youth Hymnal** *Hugo, HTML, CSS, Javascript* github.com/church-sg/cis-youth-hymnal-v2

- Open-source Progressive Web App used by 50+ church members to browse hymn lyrics with offline support.
- Built with Hugo SSG and served by GitHub Pages for fast static page loads, while Service Workers provide offline functionality.
- Wrote custom JavaScript and CSS to enable features not built into Hugo, e.g.
  - Implemented custom navigation with HTMX, allowing for seamless navigation between pages by replacing full-page browser reloads with partial DOM updates.
  - Large "previous" and "next" buttons for ergonomic browsing on mobile
  - Auto-scrolling to prior location upon returning to list view to reduce repetitive scrolling
  - Page Indexing powered by Lunr to allow for searching both title and content, increasing search accuracy and speed.
- Wrote a Bash script for a 1-click build-and-deploy onto GitHub Pages
- Wrote documentation to onboard new developers quickly to add new hymns using Markdown.

**Dotfiles** *Bash, Lua* https://github.com/notyumin/dotfiles/

- Personal Linux development environment managed with GNU Stow, including:
  - *GNU Stow*: Allows 1-command symlink of all config files to the right locations, enabling quick, reproducible workstation setup
  - *Neovim*: Built on the MiniMax distro, with custom keybindings and plugins to improve productivity and ergonomics
  - *Tmux*: Custom keybindings and minimal status bar
  - *Wezterm*: Custom colourscheme and UI tweaks

## Education

**Singapore University of Technology and Design** Singapore
BEng. Computer Science *GPA: 4.59/5* 2024 - 2028

**School of InfoComm Technology - Ngee Ann Polytechnic** Singapore
Diploma in Information Technology *GPA: 3.73/4* 2019 - 2022

**School of Science and Technology, Singapore** Singapore
O-Levels 2015 - 2018

## Awards

- SUTD Freshmore Honours List (2024/2025)
- LeafMeBe: Top in Cohort for Innovativeness and Risk-taking (2025)